

12. Dopamine is a neurotransmitter that activates the reward pathway in the brain.

(a) State one effect of activating the reward pathway.

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(b) State one method of removal of dopamine from a synapse.

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(c) (i) A dopamine agonist can be used in the treatment of Parkinson's disease. State the effect of prolonged use of an agonist.

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(ii) In a clinical trial of a potential new drug for the treatment of Parkinson's disease, individuals were divided into groups in a randomised way.

Name one other protocol that could be used during this clinical trial and explain its importance to the results.

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Protocol _____

Importance _____

(iii) State one other design factor that would allow the results of the clinical trial to have statistical significance.

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